A Digital Twin-Based State Estimation Framework for Decision Support in Primary Crushing Mining Operations

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***Abstract*—Modern mining operations increasingly rely on digital twins to support operational decision-making, yet most proposed architectures implicitly assume that the sensor and SCADA data feeding them are clean and complete. In primary crushing stations this assumption fails: severe mechanical vibration and abrasive airborne dust degrade level telemetry, and operationally relevant variables such as the instantaneous discharge rate of the crusher are not instrumented at all. This paper proposes and validates a decoupled analytical software module that acts as a physical-consistency layer between degraded telemetry and the decision layer of a mining digital twin. The module couples a mass-conservation state-space model of the crusher feed hopper with a discrete linear Kalman filter, and is validated through Monte Carlo simulation with a fixed pseudorandom seed. Over $M = 50$ independent runs of $N = 1000$ steps, the module reduces the mean squared error of the retained volume by 89.57 % with respect to the raw SCADA baseline, a reduction that is statistically significant under a paired t-test and confirmed by a Wilcoxon signed-rank test after the normality assumption is empirically rejected. More notably, for the unsensed filling rate the module reduces the error by 98.17 % relative to finite differencing of the raw signal, which indicates that the analytical layer does not merely denoise available data but generates observability where instrumentation provides none. A covariance mismatch analysis shows that performance is governed by the Q/R ratio rather than by absolute values, which substantially lowers the practical tuning barrier for industrial adoption.**

***Index Terms*—digital twin, state estimation, Kalman filter, partial observability, mining operations, decision support, cyber-physical systems, SCADA**

I. Introduction

The mining sector is undergoing a rapid digital transformation driven by the need to improve productivity, safety, energy efficiency and operational resilience. In this context, digital twins, industrial simulation and artificial intelligence have emerged as promising tools for supporting operational decision-making [1]–[3].

However, many AI- and simulation-driven systems assume that the operational state of the process is fully known. In practice, mining systems are only partially observable: measurements are noisy, delayed or incomplete, and several variables of interest are not directly sensed. This creates a gap between the physical layer and the digital decision layer.

Primary crushing is the foundational link of the copper comminution chain and illustrates the problem sharply. Extraction trucks discharge ore discontinuously into a feed hopper while the crusher extracts material continuously. The retained volume governs process continuity: too low a level produces idle operation and energy inefficiency, while too high a level leads to blockages and unscheduled downtime. Supervision relies on ultrasonic level transducers whose signals are degraded by the continuous mechanical vibration and the high concentration of abrasive dust characteristic of the crushing station. The instantaneous extraction rate, which varies with ore hardness, has no direct instrumentation at all.

Traditional automation processes this raw telemetry using deterministic algorithms or fixed-threshold alarms, which fail to capture stochastic process disturbances. When corrupted SCADA data are fed directly into higher-level optimization tools, automated decision-making degrades into suboptimal control strategies. This work addresses that analytical gap by designing a decoupled software module that performs recursive state estimation and delivers a physically consistent state to the digital twin.

The contribution of this work is threefold:

1) a dynamic-systems formulation of the primary crushing feed hopper suitable for digital twin development, in which the unsensed extraction rate is modeled as a latent state and formal observability is established;
2) a decoupled state estimation layer that fuses SCADA observations with the mass-conservation process model, validated by Monte Carlo simulation with formal hypothesis testing and an explicit verification of its distributional assumptions; and
3) a quantitative characterization of the module's robustness to sensor degradation and to covariance mistuning, which establishes the operating envelope within which the module remains useful.

The remainder of the paper is organized as follows. Section II reviews related work. Section III formulates the problem. Section IV describes the methodology. Section V reports the results, Section VI discusses them, and Section VII

concludes.

II. Related Work

A. Digital Twins and Cyber-Physical Systems

The cyber-physical systems paradigm was formulated by Lee [4] as the tight integration of computational and physical processes, noting that the central challenge is the fidelity of the coupling rather than computation itself. Uhlemann *et al.* [5] proposed the digital twin as a concrete realization of the cyber-physical production system for Industry 4.0, and Tao *et al.* [6] oriented it towards prognostics and health management of complex equipment. Fuller *et al.* [1] systematized enabling technologies and open challenges, and Liu *et al.* [2] consolidated a review of concepts, technologies and industrial applications.

A shared omission emerges from these works: all of them tacitly assume a sufficiently clean telemetry network. Uhlemann *et al.* [5] locate the data gap at the middleware and interoperability layer, while [6] and [1] focus on architecture and synchronization rather than on input signal fidelity. Sepasgozar [7] names this tension most precisely by distinguishing the *digital shadow*, which mirrors raw data, from the *digital twin* proper, which processes and removes the underlying uncertainty. This distinction is adopted here as a design criterion: without an explicit mathematical denoising layer, what is built is not a digital twin but a high-resolution digital shadow.

B. Digital Twins in the Minerals Industry

Qu *et al.* [3] provide a comprehensive review of digital twins in the minerals industry and report growing but fragmented adoption, concentrated in planning and haulage rather than in comminution stages. The more recent systematic review by Ebad *et al.* [8], covering 68 studies published between 2015 and 2025 on AI-driven digital twins in mining, reports a marked predominance of machine learning and deep learning methods and identifies data integration, real-time processing and limited industrial validation as persistent challenges.

Read jointly, both reviews support a relevant observation: research effort in mining digital twins has concentrated on the upper learning and prediction layers, while the intermediate physical-consistency layer remains comparatively unattended.

C. State Estimation in Mineral Processing

The formal framework for resolving partial observability comes from optimal estimation theory. Kalman [9] established the minimum-variance recursive estimator for linear systems in white Gaussian noise, and Simon [10] consolidated the modern treatment, including nonlinear extensions and the asymptotic convergence conditions of the error covariance.

Applications to mineral processing have solid precedents. Apelt *et al.* [11] developed inferential measurement of semi-autogenous mill parameters through state estimation, showing that non-instrumented internal variables of comminution processes can be reconstructed from indirect measurements. More recently, Bárzaga-Martell *et al.* [12] applied extended and unscented Kalman filters to the estimation of copper grade, acid consumption and moisture content in heap leaching, a work of particular relevance given its focus on copper mining.

These two bodies of literature have developed with limited contact. Work on state estimation in mining is formulated as a process control or inferential measurement problem, whereas digital twin work treats data quality as a prior assumption rather than as an architectural component to be designed. The gap addressed here is therefore not the absence of the algorithm, which has been established for over six decades, but the absence of its formulation as a software architectural component with a delimited scope, formal statistical validation and reproducibility evidence, within a mining digital twin architecture.

III. Problem Formulation

Consider the crusher feed hopper as a partially observable dynamic system. Let $\mathbf{x}_k$ denote the internal state of the process at time step k, u_k the known operational input, and y_k the measured output from the SCADA infrastructure.

The unit of analysis is the cyber-physical reception subsystem, represented by the two-dimensional discrete-time state vector

$$\mathbf{x}_k = \begin{bmatrix} V_k \\ \dot{V}_k \end{bmatrix} \in \mathbb{R}^2, \tag{1}$$

where V_k [m^3] is the ore volume retained in the crushing chamber and $\dot{V}_k$ [m^3/s] is the net instantaneous rate of volumetric change, dominated by the continuous extraction of the crusher.

The dynamics express the mass conservation of the hopper discretized by the Euler method:

$$V_{k+1} = V_k + \Delta t\, \dot{V}_k + \Delta t\, u_k, \tag{2}$$

$$\dot{V}_{k+1} = \dot{V}_k + w_k, \tag{3}$$

$$y_k = V_k + v_k, \tag{4}$$

which corresponds to the system matrices

$$\mathbf{A} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \Delta t \\ 0 \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \tag{5}$$

with $w_k \sim \mathcal{N}(0, Q_{22})$ and $v_k \sim \mathcal{N}(0, R)$.

The input u_k [m^3/s] is the truck discharge flow: a known operational event registered by the SCADA, and therefore a control input rather than a disturbance. In contrast, $\dot{V}_k$ represents the *unsensed* component of the dynamics and is modeled as a random walk. The structure of $\mathbf{H}$ makes the asymmetry explicit: the sensor observes volume and never the rate.

Reconstruction of $\dot{V}_k$ from the sequence of volume measurements is possible only if the pair $(\mathbf{A}, \mathbf{H})$ is observable. The observability matrix of the system is

$$\mathcal{O} = \begin{bmatrix} \mathbf{H} \\ \mathbf{HA} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & \Delta t \end{bmatrix}, \quad \det(\mathcal{O}) = \Delta t \neq 0, \tag{6}$$

TABLE I
EXPERIMENT PARAMETERIZATION

Parameter	Value	Rationale
Δt	1.0 s	SCADA sampling at 1 Hz
N	1000 steps	16.7 min operating horizon
M	50 runs	Monte Carlo design
Seed	42	Deterministic reproducibility
Q_{11}	0.01 m^6	Process noise on volume
Q_{22}	1.0×10^{-4}	Rate variability from ore hardness
R	5.0 m^6	Sensor noise ($\sigma_v = 2.236$ m^3)
V_0	15.0 m^3	Near-empty hopper at shift start
$\dot{V}_0$	-0.8 m^3/s	Nominal extraction rate
$\hat{\mathbf{x}}_0$	$[15.0,\ 0.0]^\mathsf{T}$	Filter does not know the rate
$\mathbf{P}_0$	$\mathrm{diag}(10.0,\ 1.0)$	Initial uncertainty
u (truck)	4.5 m^3/s	≈ 202 m^3 per 45 s event

so $\mathrm{rank}(\mathcal{O}) = 2$ for any nonzero discretization step. The system is therefore completely observable: the filling rate is mathematically reconstructible despite never being measured directly. This condition is verified computationally at the start of every run.

IV. METHODOLOGY

A. Research Design

The study adopts a quantitative, experimental design based on stochastic computational simulation. This choice is a consequence of the research question rather than a stylistic preference: quantifying an estimation error requires a ground truth against which to measure it, and in an operating plant the true internal volume is precisely what is unknown. Simulation resolves this epistemological constraint by generating the true trajectory synthetically.

B. Synthetic Data Generation and NDA Mitigation

Non-disclosure agreements prevalent in the mining industry restrict access to historical raw SCADA telemetry. The study therefore uses parametric synthetic data: a truth model based on flow conservation laws, into which white Gaussian process and measurement noise are injected. The covariance matrices are parameterized within operating ranges consistent with the physics of the asset.

Table I summarizes the parameterization. A nominal extraction of 0.8 m^3/s corresponds to approximately 5760 t/h at a bulk density of 2.0 t/m^3, consistent with a large-capacity gyratory crusher. Each truck discharge contributes about 202 m^3, consistent with a 300–400 t haul truck. Four truck cycles over the simulated horizon keep the hopper within an operating band of 7 to 300 m^3.

It is worth noting that the process noise on the rate was recalibrated during implementation. An earlier design value of $Q_{22} = 0.05$ produces, over $N = 1000$ steps, a random walk with accumulated standard deviation $\sqrt{1000 \times 0.05} = 7.07$ m^3/s, nearly nine times the nominal extraction rate. Such a regime is physically inadmissible for the asset. The recalibrated value yields an accumulated dispersion of 0.32 m^3/s around the nominal rate, consistent with variability attributable to ore hardness.

C. State Estimation Layer

A discrete linear Kalman filter combines the process model with the sensor observations through the standard predictor–corrector cycle:

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{A}\hat{\mathbf{x}}_{k-1|k-1} + \mathbf{B}u_k, \tag{7}$$

$$\mathbf{P}_{k|k-1} = \mathbf{A}\mathbf{P}_{k-1|k-1}\mathbf{A}^\mathsf{T} + \mathbf{Q}, \tag{8}$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}^\mathsf{T}\left(\mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^\mathsf{T} + \mathbf{R}\right)^{-1}, \tag{9}$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k\left(y_k - \mathbf{H}\hat{\mathbf{x}}_{k|k-1}\right). \tag{10}$$

The covariance update uses the Joseph form,

$$\mathbf{P}_{k|k} = \mathbf{\Gamma}\mathbf{P}_{k|k-1}\mathbf{\Gamma}^\mathsf{T} + \mathbf{K}_k\mathbf{R}\mathbf{K}_k^\mathsf{T}, \quad \mathbf{\Gamma} = \mathbf{I} - \mathbf{K}_k\mathbf{H}, \tag{11}$$

which is algebraically equivalent to the simplified form but numerically stable: it preserves the symmetry and positive definiteness of $\mathbf{P}$ under floating-point rounding, a necessary condition over long horizons.

The estimator core is implemented as a generic class decoupled from the physical process: it receives the system matrices and operates without knowledge of the application domain. This property was verified empirically by applying the same unmodified core to three additional subsystems of the asset — motor active power, stator thermal dynamics and liner wear — by substituting only the system matrices.

D. Evaluation Metrics and Statistical Tests

Performance is evaluated against the raw SCADA reading as the declared baseline, using the mean squared error and its root:

$$\mathrm{MSE} = \frac{1}{N}\sum_{k=1}^{N}\left(V_k - \tilde{V}_k\right)^2, \quad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}, \tag{12}$$

where $\tilde{V}_k$ denotes either y_k or $\hat{V}_{k|k}$. Since volume is expressed in m^3, the MSE is necessarily expressed in m^6, whereas its root recovers the units of the physical variable. All statistical inference is therefore performed on the RMSE, so that test statistics, confidence intervals and effect sizes are dimensionally homogeneous with the estimated state.

Each Monte Carlo run is an independent realization and contributes one pair $(\mathrm{RMSE}_{\mathrm{SCADA},m}, \mathrm{RMSE}_{\mathrm{KF},m})$. Over the M pairs, with d_m their difference, we test

$$H_0 : \mu_d = 0 \quad \text{against} \quad H_1 : \mu_d > 0, \quad \alpha = 0.05, \tag{13}$$

using a one-sided paired t-test. The normality assumption of the differences is not assumed but verified through the Shapiro–Wilk test, and a Wilcoxon signed-rank test is additionally reported as a robustness contrast so that the conclusion does not depend on the parametric assumption holding.

TABLE II
ERROR METRICS FOR VOLUME RECONSTRUCTION ($M = 50$, $N = 1000$)

Metric	Raw SCADA	Kalman filter	Reduction
MSE [m^6]	4.9660	0.5179	89.57 %
MSE std. dev.	0.2340	0.0865	—
RMSE [m^3]	2.2278	0.7173	67.80 %
RMSE std. dev.	0.0522	0.0588	—

TABLE III
ERROR METRICS FOR THE UNSENSED FILLING RATE

Metric	Finite diff.	Kalman filter	Reduction
RMSE [m^3/s]	3.6891	0.0674	98.17 %

For the latent state, the baseline is finite differencing of the raw signal, $\dot{V}_k^{\mathrm{fd}} = (y_k - y_{k-1})/\Delta t$, which is the only procedure available to a conventional SCADA system attempting to estimate the filling rate without a state estimator.

E. Sensitivity Analysis

Two complementary experiments characterize the operating envelope. A *degradation sweep* increases the true measurement noise variance from benign to severe conditions with the filter correctly tuned. A *mismatch grid* tunes the filter with covariances scaled by factors (α, β) with respect to the true ones, so that $\mathbf{Q}_{\text{filter}} = \alpha\,\mathbf{Q}$ and $\mathbf{R}_{\text{filter}} = \beta\,\mathbf{R}$, representing the realistic scenario in which a plant engineer calibrates the estimator with imprecise parameters. All combinations are evaluated on identical physical scenarios so that observed differences are attributable solely to calibration.

V. RESULTS

All reported values come from executing the module with the parameterization of Table I and a fixed seed, and are reproducible.

A. Volume Reconstruction

Table II reports the error metrics on the observed variable, averaged over $M = 50$ independent runs.

The MSE of the raw telemetry, 4.97 m^6, matches the injected noise variance $R = 5.0$ m^6 by construction, which serves as an internal verification of the degradation generator. Fig. 1 shows the reconstruction over a representative run; the sawtooth profile corresponds to the real operational cycle of discontinuous truck discharge over continuous extraction.

B. Latent Variable Recovery

Table III reports the results on the filling rate, which has no instrumentation of any kind.

The magnitude of the difference deserves attention. Finite differencing of the raw signal produces an error of 3.69 m^3/s over a physical quantity whose nominal value is 0.8 m^3/s: the procedure does not return an imprecise estimate, it returns noise with the appearance of a measurement. Fig. 2 contrasts both procedures; note the difference of scale between panels.

TABLE IV
PAIRED STATISTICAL INFERENCE ON VOLUME RMSE ($M = 50$, $\alpha = 0.05$)

Statistic	Value
Mean RMSE reduction	1.5106 m^3
95 % confidence interval	$[1.4881,\ 1.5330]$ m^3
Paired t-test (one-sided)	$t(49) = 135.24$; $p = 5.15 \times 10^{-65}$
Effect size (Cohen's d)	19.13
Shapiro–Wilk normality	$W = 0.9459$; $p = 0.0233$ (rejected)
Wilcoxon signed-rank	$W = 1275.0$; $p = 8.88 \times 10^{-16}$
Decision	H_0 rejected

TABLE V
MODULE PERFORMANCE UNDER INCREASING INSTRUMENTAL DEGRADATION

R [m^6]	σ_v [m^3]	MSE SCADA [m^6]	MSE KF [m^6]	Reduction
1	1.00	0.987	0.165	83.32 %
5	2.24	4.937	0.527	89.33 %
20	4.47	19.748	1.449	92.66 %
50	7.07	49.369	2.831	94.27 %
100	10.00	98.738	4.707	95.23 %
200	14.14	197.476	7.843	96.03 %
400	20.00	394.953	13.093	96.68 %

C. Asymptotic Convergence

Fig. 3 shows the convergence of the error covariance and of the Kalman gain towards their steady-state values, the behavior predicted by the Riccati formulation for linear time-invariant systems [10]. Convergence is reached at around 60 s of operation, with $K_1 \to 0.0993$.

D. Statistical Validation

Table IV reports the inference on the RMSE.

The normality verification merits comment. The Shapiro–Wilk test *rejects* normality of the differences ($p = 0.0233 < 0.05$), so a mechanical application of the t-test would be questionable. It is precisely for this reason that the design incorporated the nonparametric Wilcoxon contrast, which requires no normality and confirms the conclusion with $p = 8.88 \times 10^{-16}$. The agreement of both tests allows H_0 to be rejected independently of the parametric assumption. It should be added that with $M = 50$ observations the central limit theorem grants the t-test considerable robustness to moderate departures from normality, and that the effect size — a Cohen's d of 19.13 — places the result well beyond any distributional consideration. For the latent state, inference yields $t(49) = 296.36$ with $p = 1.09 \times 10^{-81}$ and a 95 % confidence interval of $[3.5971,\ 3.6463]$ m^3/s.

E. Sensitivity to Sensor Degradation

Table V and Fig. 5 report the behavior of the module under increasing levels of true sensor noise.

Error reduction grows monotonically with the severity of degradation, from 83.32 % under benign conditions to 96.68 % under extreme degradation. The contribution of the module

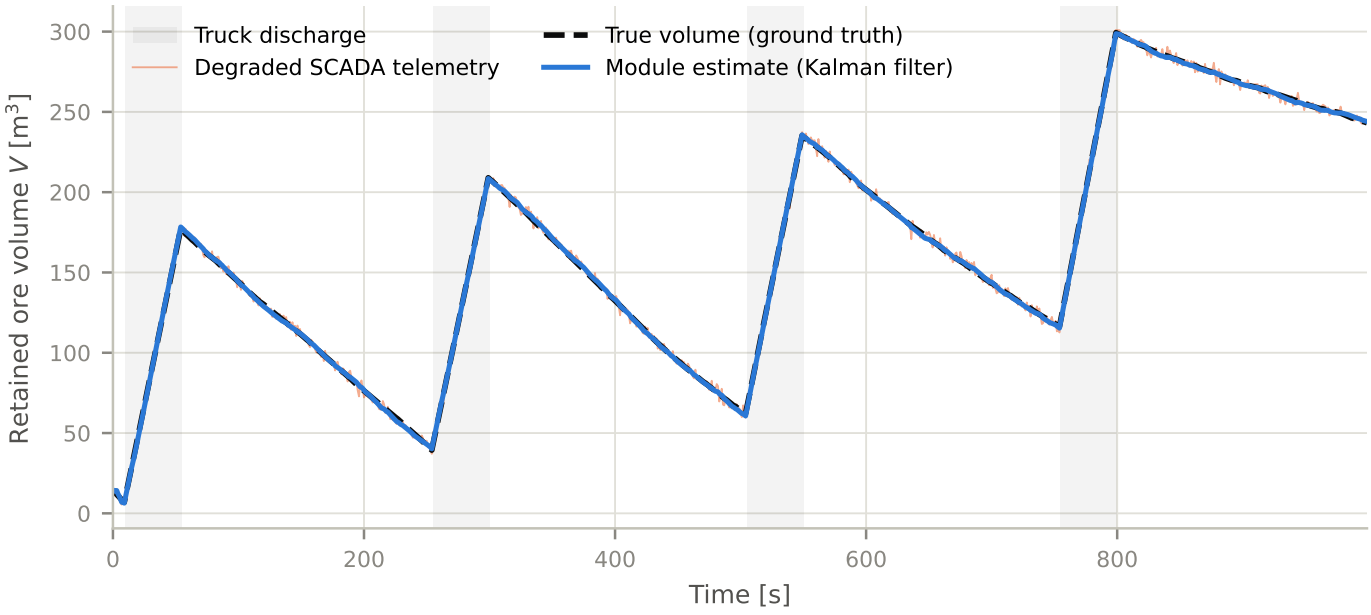


Fig. 1. Dynamic reconstruction of the retained ore volume in the feed hopper. Shaded bands indicate truck discharge events.

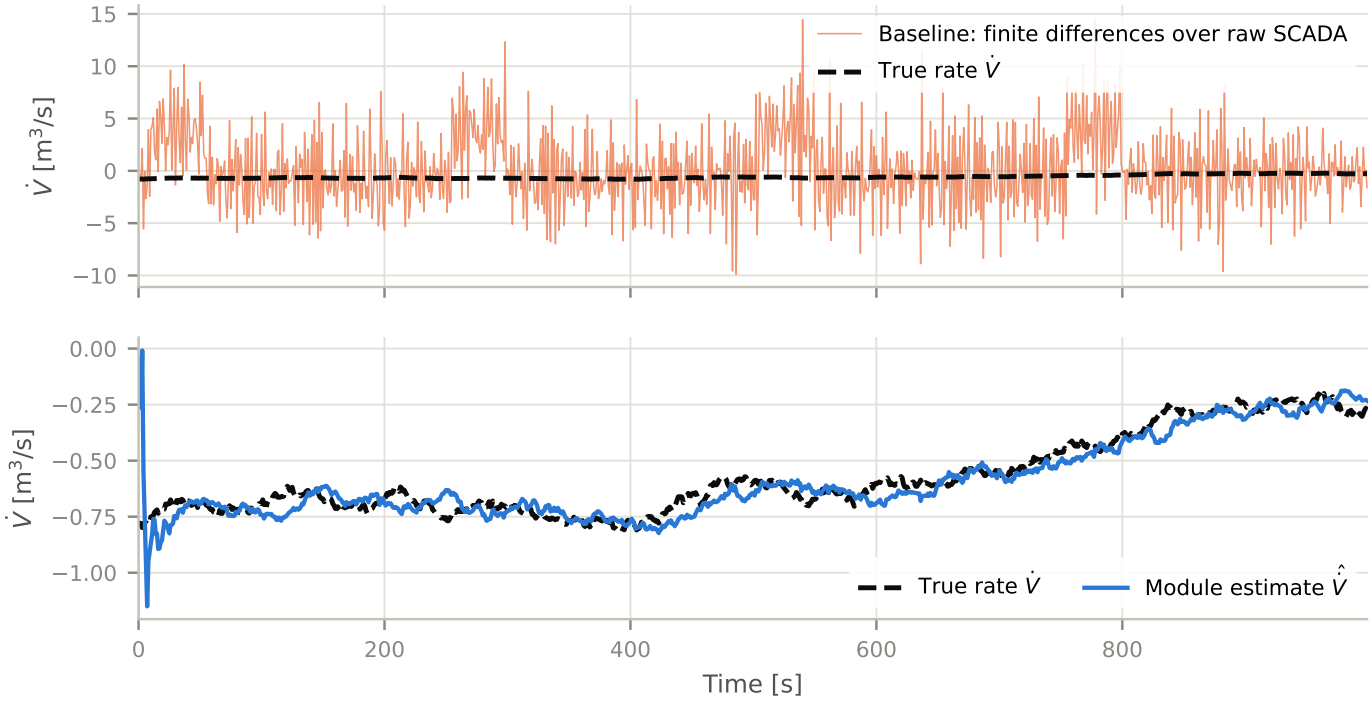


Fig. 2. Recovery of the latent variable. Top: finite-difference baseline over the raw signal. Bottom: module estimate, at physical scale.

therefore does not dilute in hostile environments: it becomes more pronounced.

F. Robustness to Covariance Mismatch

Fig. 6 presents the RMSE grid resulting from deliberately mistuning the filter covariances with respect to the true physics.

The result exhibits a marked diagonal structure. Under nominal tuning the RMSE is 0.721 m^3, against 2.224 m^3 for the raw telemetry. Along the diagonal, where both covariances are scaled by the same factor even up to a hundredfold in either direction, performance remains in the range 0.72–0.85 m^3. Severe degradation appears only in the anti-diagonal corners, with a worst case of 8.45 m^3. Of the twenty-five combinations evaluated, only three are worse than not filtering at all.

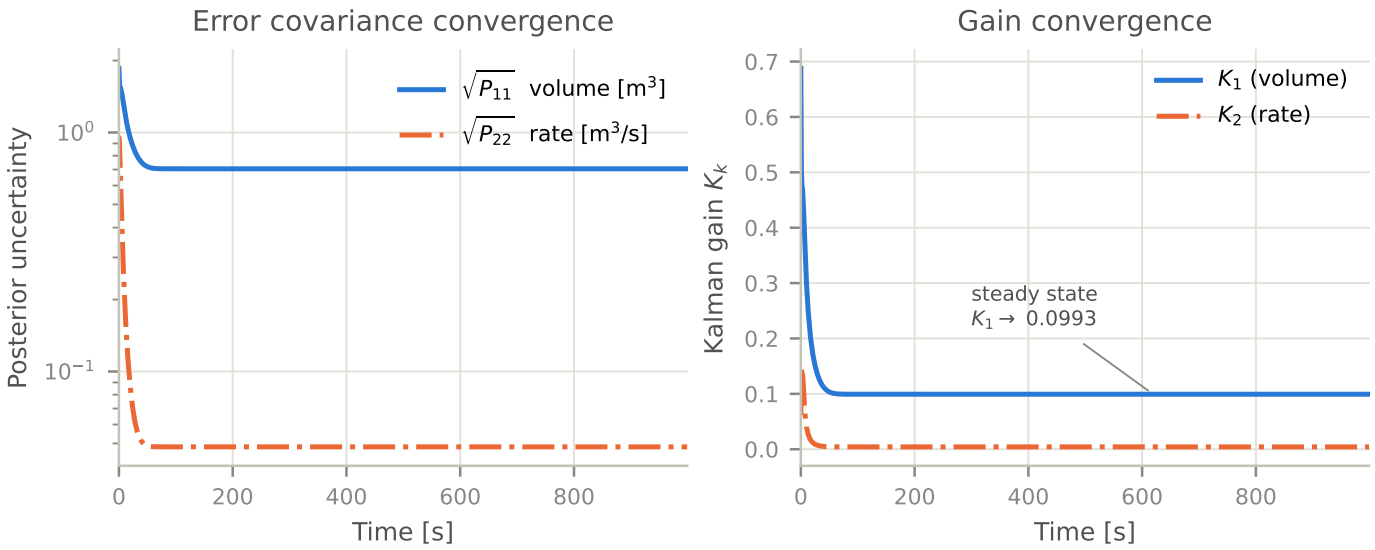


Fig. 3. Asymptotic convergence of the estimation error covariance and of the Kalman gain towards steady state.

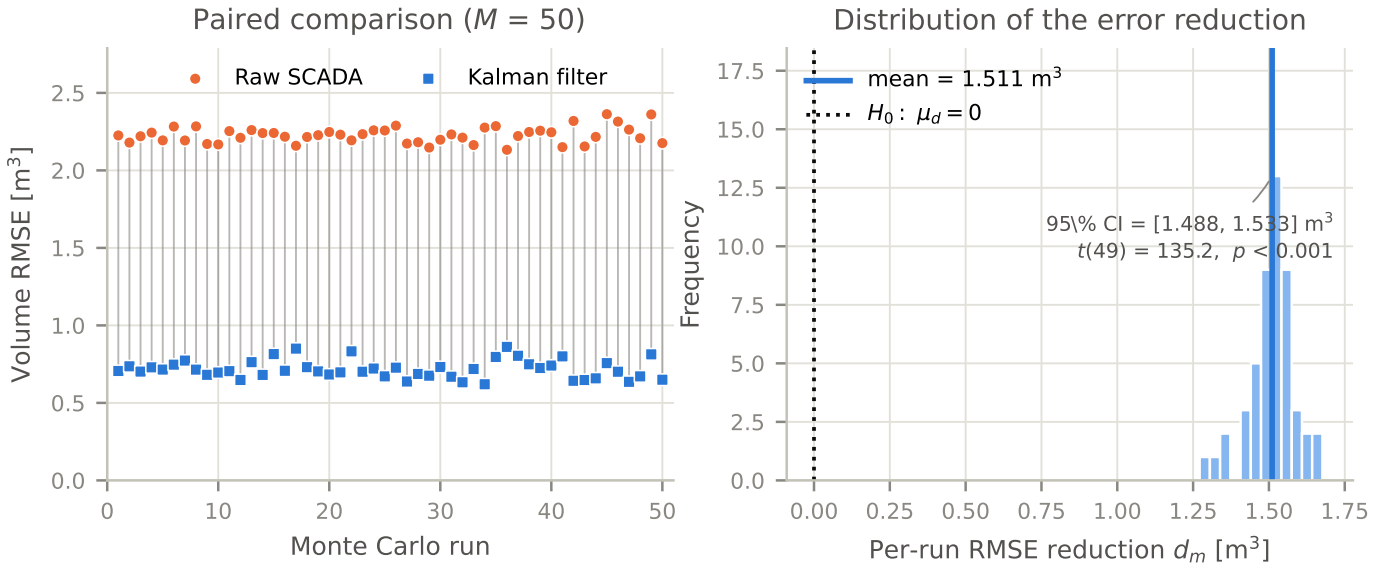


Fig. 4. Monte Carlo validation. Left: run-by-run paired comparison. Right: distribution of the RMSE reduction, with the null hypothesis marked at the origin.

G. Reproducibility

Two successive independent executions of the experiment produced byte-identical output files, verified by MD5 checksum over the four generated logs. Exact reproducibility of the study is therefore a verified fact rather than a declaration of intent.

VI. Discussion

A. Creating Observability Rather Than Denoising

The most conceptually dense result is not the error reduction on volume but the reconstruction of the filling rate. On volume, the module improves a signal that *already exists*: a sensor measures it and the filter reduces its variance. That is a valuable but incremental contribution. On the filling rate, the module produces information that *did not exist anywhere in the system*. No instrument measures it, and the only procedure available to a conventional SCADA returns noise.

That the module reduces this error by 98.17 % means that the analytical layer is not cleaning data but *creating observability* where instrumentation provides none. This is the strongest argument for conceiving the estimator as an architectural component rather than as a preprocessing filter: a preprocessor improves what it receives, whereas this module delivers to the upper layers a state variable that was previously unavailable for decision-making.

B. Relation to Prior Work

The findings give quantitative support to the conceptual distinction of Sepasgozar [7]: if the raw telemetry of the filling

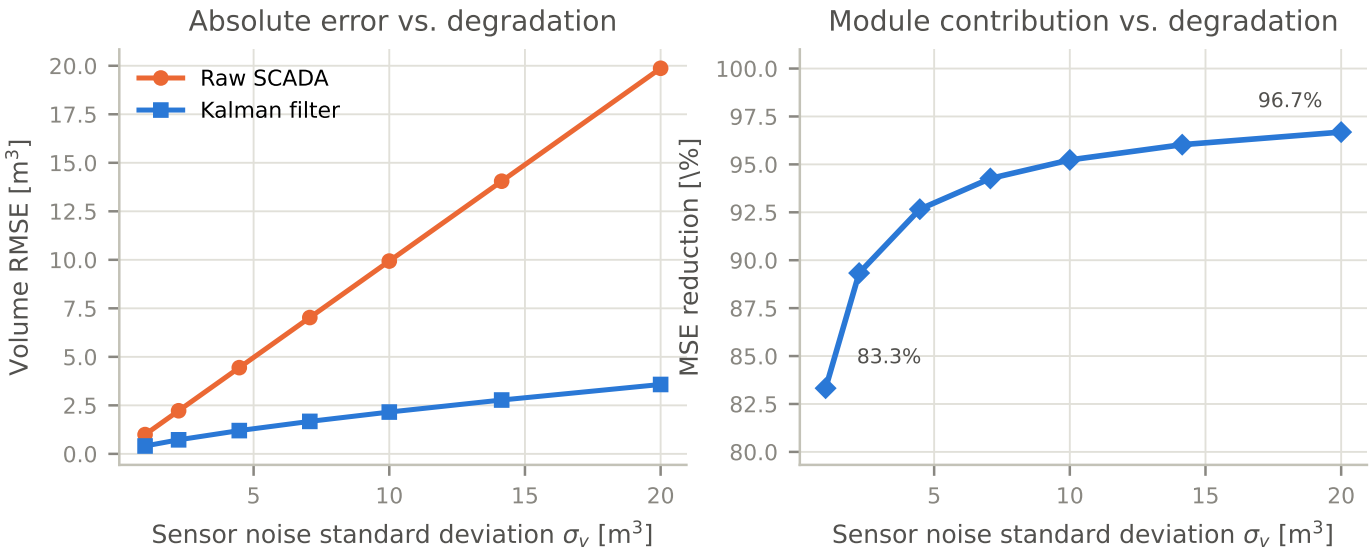


Fig. 5. Module behavior under increasing instrumental noise. Left: absolute error of both sources. Right: relative MSE reduction.

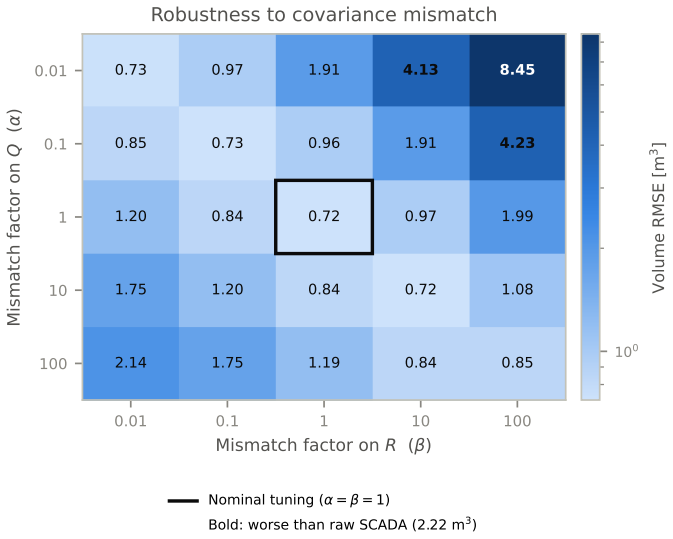


Fig. 6. Robustness of the module to deliberate mismatch of the $\mathbf{Q}$ and $\mathbf{R}$ covariances. The box marks the nominal tuning.

rate is essentially noise, a virtual replica fed directly by the SCADA is not representing the asset but its distortion. The difference between digital shadow and digital twin ceases to be terminological and acquires a measurable magnitude.

With respect to mining digital twins, the study aligns with the diagnosis of Ebad *et al.* [8] regarding limited industrial validation and persistent data integration problems, and with the observation of Qu *et al.* [3] on fragmented adoption. The contribution addresses precisely the layer both reviews identify as unattended.

With respect to state estimation in mineral processing, the results are consistent with Apelt *et al.* [11] in semi-autogenous grinding and with Bárzaga-Martell *et al.* [12] in copper heap leaching: non-instrumented internal variables of mining processes are effectively reconstructible. The difference of the present work is not one of result but of framing. Those studies formulate estimation as an inferential measurement or process control instrument; this work formulates it as a component of a digital twin architecture, with the decoupling, reproducibility and statistical validation requirements that framing imposes. It is worth noting that [12] employs nonlinear variants while this work deliberately restricts itself to the linear case: the mass balance of a hopper is genuinely linear over the operating range considered, and a nonlinear estimator would add computational cost without precision gain.

C. Practical Implication of the Robustness Finding

The diagonal structure of the mismatch grid shows that estimator performance is governed by the *ratio* between covariances rather than by their absolute values. A calibration error of two orders of magnitude in both matrices simultaneously degrades the RMSE only from 0.72 to 0.85 m^3, whereas the same error applied to a single one raises it to 8.45 m^3.

The implication for industrial deployment is direct. The practical difficulty of tuning a Kalman filter in a plant does not lie in determining the absolute values of $\mathbf{Q}$ and $\mathbf{R}$, which would require detailed metrological characterization, but in correctly estimating their proportion, that is, how much to trust the physical model relative to the sensor. This is a considerably more accessible task for a process engineer and lowers the adoption barrier of the module.

D. Limitations

The principal limitation is that validation is performed on synthetic data generated by the same model the filter uses. Under that condition the Kalman filter is optimal by theoretical construction, so the study verifies correct implementation and quantifies the magnitude of the improvement, but does not subject the estimator to a genuine model discrepancy. The

covariance mismatch analysis partially mitigates this objection without eliminating it: a real process may exhibit nonlinearities, correlated or colored noise, and unmodeled drifts that this design does not reproduce.

A second limitation concerns the scope of the physical model. The implemented mass balance does not incorporate the real chamber geometry, material size segregation, or the dependence of extraction rate on instantaneous ore hardness. Start-up and shutdown transients and blockage events were excluded, which are precisely the conditions of greatest operational interest.

A third limitation is the absence of validation with real telemetry, a direct consequence of confidentiality restrictions. This places the work at a laboratory and simulation technology readiness level, and none of the reported figures should be interpreted as a prediction of plant performance. Any deployment on an operating digital twin requires prior on-site piloting and empirical safety calibration.

VII. Conclusion and Future Work

This paper presented and validated a decoupled analytical software module that acts as a physical-consistency layer between degraded SCADA telemetry and the decision layer of a mining digital twin. Over 50 independent Monte Carlo runs, the module reduced the mean squared error of the retained volume by 89.57 %, a statistically significant reduction robust to the distributional assumption, and reduced the error on the unsensed filling rate by 98.17 %.

The contribution does not lie in the algorithm, established theory since [9], but in three distinct planes: architecturally, the study formulates stochastic state estimation as a design component of a mining digital twin rather than as a control instrument or preprocessing stage; empirically, it provides quantified evidence that the difference between a digital shadow and a digital twin is measurable, and that the analytical layer can generate observability where instrumentation provides none; and methodologically, it delivers a byte-reproducible artifact with traceability logs, verification of statistical assumptions and parametric robustness analysis, in an area where effective validation has been reported as a shortcoming.

Four continuation lines are identified. The most immediate is validation with real telemetry from an operating site, which would allow performance to be contrasted under genuine model discrepancy and is a necessary condition for any deployment. The second is extension to nonlinear regimes, incorporating the start-up, shutdown and blockage conditions excluded here, which would require migrating towards extended or unscented variants following [12]. The third is adaptive covariance tuning, suggested directly by the $\mathbf{Q}/\mathbf{R}$ ratio finding: a mechanism adjusting that proportion online from the innovation sequence would further reduce the industrial adoption barrier. The fourth is generalization of the module as a transversal service of the digital twin, applying the same core to the different subsystems of the asset.

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